LESSON PLAN

AIRCRAFT ARMAMENT (18)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	12
Total No. of Lesson Plans	12	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Basic concept of ECCM
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Basic concept of ECCM. Various methods of ECCM employed by Indian Naval aircraft viz. frequency agility PRF jittering etc
5. Specific Objectives	18.1.4.1 Trainee will be able to appraise the function of secondary radars in ASW operation.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of ECCM diagrams in IB; Instructor supervision required for diagram interpretation
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a radar system?
(b) Define the term Electronic Counter-Counter Measures (ECCM).
(c) Why is ECCM necessary in modern naval warfare?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition and basic concept of ECCM • Distinction between ECM and ECCM • Threat environment for naval aircraft • Role of ECCM in ASW operations • Importance of secondary radar integration	L & Ds	PPT, IB	15 min
(b)	• Frequency agility techniques • Pulse Repetition Frequency (PRF) jittering • Dwell time management • Implementation in Indian Naval aircraft • Effectiveness against modern threats	L & Ds	PPT, PDF	20 min
(c)	• Secondary radar function in ASW • Data link integration with ECCM • Limitations of secondary radar systems • Coordination with primary radar • Operational scenarios for ASW	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how frequency agility disrupts enemy radar tracking.
(b) Describe the role of PRF jittering in enhancing aircraft survivability.
(c) How does the secondary radar support ASW operations alongside ECCM?

12. Home Assignment and Follow Up	Prepare a brief note on how the concepts of ECCM discussed in this lesson will be applied in the next lesson plan regarding advanced radar systems.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	End of subject	Refer to the master syllabus reference list	-

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min